



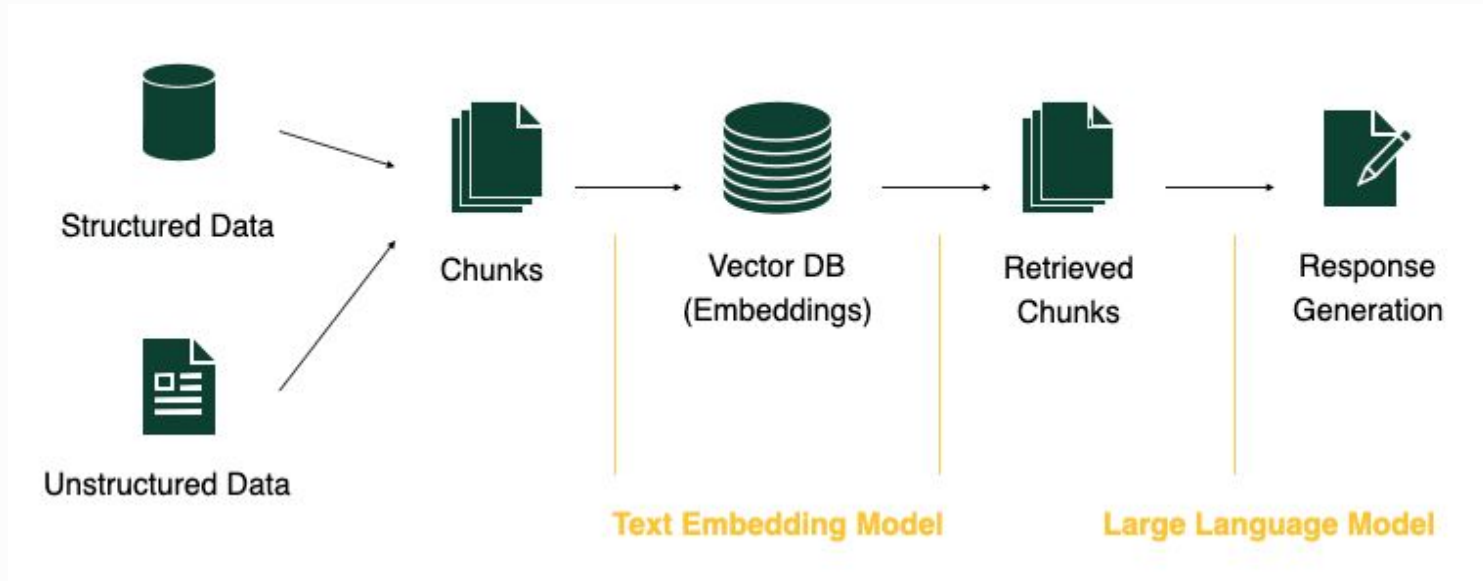
Real World RAG System

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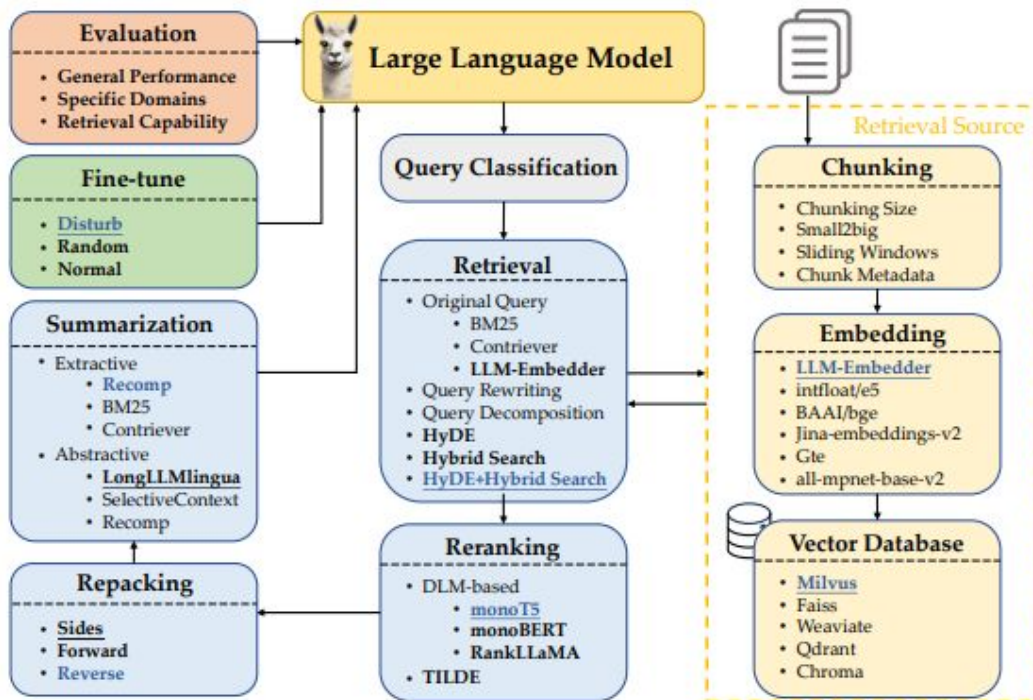
Mentors : Gopichand, Lokesh



Simple RAG Workflow



Simple RAG vs Advanced RAG



Evaluation

- RAG systems are prone to hallucinations as the generator model struggles to retrieve relevant information from the context.
- Tuning an optimal system for a particular RAG application involves iterative evaluation of multiple configurations.
- Are there any benchmark datasets for evaluation?
 - RAGBench
 - RGB

RAGBench Dataset

- Large-scale benchmark dataset containing 100k examples spanning five industry-specific domains.
 - Biomedical Research
 - General Knowledge
 - Legal
 - Customer Support
 - Finance
- Link : <https://huggingface.co/datasets/rungalileo/ragbench>

RAGBench - Key Evaluation Metrics

1. **Context Relevance:** *fraction of the retrieved context that is relevant to the query*
2. **Context Utilization:** *fraction of the retrieved context that is used by the generator to produce the response*
3. **Completeness:** *how well the response incorporates all the relevant information in the context*
4. **Adherence:** *whether or not all parts of the response are grounded in the context*

RGB Dataset

Analyze the generation capabilities of LLMs in 4 fundamental abilities required for RAG.

1. **Noise Robustness:** *Ability to produce correct answer in the presence of noisy information*
2. **Negative Rejection:** *Ability to reject answering when there is insufficient info*
3. **Information Integration:** *Ability to integrate info from multiple documents*
4. **Counterfactual Robustness:** *Ability to detect and correct factually incorrect info*

For more details refer: [document](#)

Bonus Components - RAGBench

1. Rigorous Evaluation for Chunking & Semantic Similarity:

- *Explore and evaluate novel improvements to chunking and semantic similarity techniques for enhancing retrieval quality in RAG systems.*

2. Domain-Specific Training for Semantic Chunking:

- *Train/adapt semantic chunking methods for domain-specific datasets.*
- *Evaluate performance improvements over generic chunking approaches.*

Bonus Components - RGB

Model Capability Improvements with the same Base Model:

Focus on improving:

- Noise robustness
- Negative rejection capability
- Information Integration
- Counterfactual robustness

without changing the base LLM architecture.

Bonus Component - Enterprise RAGBench

- Investigate the feasibility of implementing the Enterprise RAGBench paper for enterprise-scale RAG evaluation. Dataset covers:
 - Slack
 - Gmail
 - HubSpot
 - Fireflies
 - GitHub
 - Jira, and Confluence
- If full implementation is not feasible, include it as an extended analysis and proposal component to RAGBench task
- Paper Link: <https://arxiv.org/pdf/2605.05253>

Task

1. Go through the following research papers and summarize your understanding.

RAGBench: <https://arxiv.org/pdf/2407.11005>

RGB: <https://arxiv.org/pdf/2309.01431>

RAG best practice: <https://aclanthology.org/2024.emnlp-main.981/>

2. Project Proposal Document (Deadline 30-May-2026)
 - a. Problem statement
 - b. Dataset Description
 - c. Methodology
 - d. Evaluation metrics
 - e. Deployment (FastAPI, Gradio)
 - f. Expected timeline



Thank You